Ruthless Controller Relay — What Works Cheat Sheet

Plain-English guide: which controllers work, how well, and why. No tech knowledge needed. For version 2.0.0.

The short version

> **Best experience: a USB cable for PlayStation controllers, a wireless dongle for Xbox controllers.** That way everything works: moving, buttons, vibration, and light color. Bluetooth works great for moving and buttons on every controller, but PlayStation controllers (DualSense / DualShock 4) lose vibration and light color over Bluetooth — see why below. This is not something wrong with your setup, it's true for everyone, everywhere.

Compatibility matrix

Controller	How it's connected	Movement & Buttons	Vibration (Rumble)	Light Color	Notes
PS5 Controller (DualSense)	**USB Cable**	**Works**	**Works**	**Works**	Everything works. Recommended.
PS5 Controller (DualSense)	**Bluetooth (wireless)**	**Works**	**Doesn't work**	**Doesn't work**	Plug in a cable if you want rumble/light.
PS4 Controller (DualShock 4)	**USB Cable**	**Works**	**Works**	**Works**	Everything works. Recommended.
PS4 Controller (DualShock 4)	**Bluetooth (wireless)**	**Works**	**Doesn't work**	**Doesn't work**	Plug in a cable if you want rumble/light.
Xbox Controller	**USB Cable**	**Works**	**Works**	n/a *	Fine for normal play. **HOTAS mode does NOT reliably work this way** (V / M unreliable too) — see warning below (†).
Xbox Controller	**Wireless Dongle (official Xbox Wireless Adapter, or the controller's own)**	**Works**	**Works**	n/a *	Fully works, HOTAS/V/M included. Recommended over a cable for this controller specifically.
HOTAS / Flight Stick / Throttle	**USB Cable**	**Works**	n/a *	n/a *	These don't have vibration motors or a color light in real life, so there's nothing to turn on.

* n/a = that controller doesn't physically have this feature, so it's not a bug or limitation of the software.

⚠† HOTAS mode does NOT reliably work with a wired Xbox controller
HOTAS mode is the main reason this program exists — and a wired Xbox controller is the one case where it can't be trusted. Windows connects a wired Xbox controller through a special driver that this software (or anything else) has no ability to hide from other apps, unlike every other controller in this list. That means the game can still notice the real controller and use normal gamepad controls instead of reading the joystick axes this program is feeding it through HOTAS mode — the mode may look like it's working on this screen while the game itself ignores it. The same underlying problem also makes **V** (disguise as PlayStation) and **M** (button remap) unreliable for this controller.

Fix: connect that Xbox controller through a wireless dongle instead of a cable (the official Xbox Wireless Adapter, or the controller's own 2.4GHz dongle if it has one) — same controller, same buttons, just a different plug — and HOTAS mode, V, and M all work exactly as expected.

Why doesn't PlayStation Bluetooth rumble/light work?

It's not broken — nobody has cracked this. When a PS4/PS5 controller connects over a USB cable, it accepts vibration and light-color commands using a simple, well-understood method. Over Bluetooth, Sony uses a different, much stricter method that basically no software outside of an actual PlayStation console has ever gotten working reliably — including big well-known controller tools that have been trying for years. We tested it thoroughly (including trying a second, different Bluetooth adapter) and confirmed the same result every serious attempt online has run into. Moving and pressing buttons over Bluetooth is completely unaffected — only vibration and the light color are impacted.

Two versions are included — which one do I run?

Version	What it does	When to use it
RuthlessControllerRelay.exe	Full version — vibration, light color, and every optional feature.	Use this one first. This is the normal, recommended version.
RuthlessControllerRelay_SAFE.exe	Stripped-down version — no window, vibration, light color, battery reading, head tracking, update check or Stream Deck control. Just clean, reliable button/stick input.	Only use this as a backup if the full version ever acts strange on your PC. It's a simpler fallback with fewer moving parts.

Controls — what the keys do

Key	What it does
`Ctrl` + `Alt` + `H` or the controller button shown on the dashboard	Switch between **Normal mode** (game sees your controller normally) and **HOTAS mode** (game sees a joystick). Do this once at the start — this is the main control. It works from anywhere.
`V`	Flip the OUTPUT badge between PlayStation-style and Xbox-style — use this if a specific game only recognizes one type.
`R`	Change which key or controller-button-combo does the Normal/HOTAS switch above.
`M`	Remap a button — make one button act like a different one.
`G`	Pick which game this launches, or launch it directly next time.
`F`	Free look: off, normal, or head tracking.
`H`	Set up head tracking, or change its camera or game.
`+` / `-`	Head tracking speed, while it's on. Saved automatically.
`Ctrl` + `Alt` + `K` or `L`	Push the look left/right or up/down axis alone for 8 seconds, so your game can map it. Menus only.
`U`	Install an update, when the title line says a new version is out. It asks first.
`C`	Turn the automatic update check on or off.

Everything is in the window: click a box at the top, or the gear at the bottom right. The letter keys are for the old console, which is still there: gear, then App, then Show console. Click the console first so it's the active window, not your game.

Looking around in HOTAS mode

On the controller	What it does
Hold **RB** (**R1** on PlayStation)	Free look: the right stick's left/right turns the camera. Let go and it snaps back.
RB + click **R3**	Lock the camera where it is. Tap RB or click R3 to unlock.
Tap **RB** (head tracking on)	Recenter. Look straight ahead first.
Hold **RB** (head tracking on)	Your head pauses and the stick has the camera, so you can look behind you. Let go to return to your head.

Map the look axes once, or RB won't move the camera. On your game's HOTAS screen, press Ctrl+Alt+K and click its look left/right row within 8 seconds, then Ctrl+Alt+L for look up/down. For the camera to snap back or recenter, bind your game's free look reset, if it has one, to RB's joystick button: button 6, which some games show as button 5.

Head tracking in one minute

Click the gear: pick your webcam under Head tracking, your game under Game, and set Look to Head. Map the look axes once (above). There's nothing to install. Then start the game, switch to HOTAS mode and turn your head. The dot in the ring shows where you're pointing. It only runs with the game you picked, and only works in games that let you bind look axes to a joystick. To use it in another game, change Game and map the axes there too.

Anti-cheat notice. Since 2.0, the relay sends no mouse movement and no keys. Your head moves joystick axes that you map on the game's own screen. One player whose head tracker moved the mouse has reported a ban for hacking, and the game studio's CEO said its detection may have seen it as inhuman mouse movement. This doesn't move the mouse, so we don't think it gets flagged that way. That's our reasoning, not a ruling: the game hasn't said head tracking is allowed, so nobody can promise it's safe. Use it at your own risk.

Look box says	Meaning
OFF	Head tracking is off, or you're in Normal mode. Switch to HOTAS mode, and set Look to Head.
WAITING	Head tracking is on, but it isn't reading a head yet. Normal until your game is running. If it's running, face the camera, check that the head_tracker folder is next to the program, then check your camera under Head tracking.
TRACKING	Your head is moving the look axes. Camera not moving? Map the look axes (above).
PAUSED	No head readings for a moment. It comes back by itself.
RB: STICK	You're holding RB, so the stick has the camera.
LOCKED	The camera is frozen with RB + R3.

Reading the screen: right vs. wrong

This is the old console (gear, then App, then Show console). The window's three boxes at the top say the same things in fewer words. In the console, the top banner and the two INPUT/OUTPUT boxes are the whole story — glance at those first any time something feels off.

✓ Everything's fine

- Banner reads **NORMAL MODE** or **HOTAS** — colored **teal**, **purple**, or **green** (never orange or red)
- **Controller:** line names your actual controller (e.g. "PlayStation 5 controller (DualSense)")
- Both INPUT and OUTPUT boxes show a real type — **PS**, **XBOX**, or **HOTAS** (never blank dashes)
- Sticks/bars/buttons on screen move the instant you touch the real controller

This is your green light — safe to start the game.

✗ Something's wrong

- **"NOT READY — waiting for a controller..."** (orange/red banner)
  Fix: turn the controller on / plug it in. It's picked up automatically — no restart needed. Don't start your game until this clears.
- **"MODE UNKNOWN — press Ctrl+Alt+H once to set it"** (red banner)
  Fix: press Ctrl+Alt+H once to choose Normal or HOTAS mode.
- OUTPUT box shows **"----"**
  Fix: no virtual controller made yet — set the mode (above) first; check the log file if it still doesn't clear.
- "This window is too short to show the dashboard..."
  Fix: maximize/enlarge the window, or shrink the font — it fixes itself automatically once there's room.
- The **Battery** line turns **red**
  Fix: the battery is low. Charge the controller.

Two more live indicators worth knowing:

| PS | | XBOX |

- The **face buttons light up their real-life color** when pressed (PlayStation Triangle/Square/Cross/Circle or Xbox Y/X/A/B) — unpressed buttons stay plain gray. If a button never lights up, that specific button isn't registering.
- **Banner color meanings in full:** teal = Normal mode, output matches your real controller type (PS→PS or Xbox→Xbox); purple = Normal mode but output is set to the other type on purpose (e.g. real PS controller showing up as Xbox — this is fine if you did it with V); green = HOTAS mode; yellow = free look, you're looking around; magenta = free look, camera locked; orange/red = mode not chosen yet or no controller found, see the table above.

Something not working?

> Every run writes a plain-text log file next to the program (**driver_install_log.txt**). If anything looks wrong, that file has the details — send it along if you're asking for help troubleshooting. There's also an **Uninstall.bat** included that cleanly removes everything the program installed, if you ever want to fully undo it. Head tracking installs nothing: it lives in the head_tracker folder beside the program, so deleting that folder removes it.
